

Saddle-Dominated Capacity of LSE Dense Associative Memory: status report

Abstract

The log-sum-exp (LSE) Dense Associative Memory [4] stores M unit vectors ξ^μ ($\mu = 1, \dots, M$) on the N -sphere of radius $\sqrt{N}$ and assigns to each chain state $\mathbf{x}$ the energy $H(\mathbf{x}) = -\beta_{\text{net}}^{-1} \ln \sum_{\mu} \exp(\beta_{\text{net}} \mathbf{x} \cdot \xi^\mu)$, with β_{net} the kernel sharpness (inverse variance). At load $\alpha = \ln M/N$ and external temperature T , the question is whether a chain at ξ^1 stays close to it.

Ramsauer et al. proved $\alpha_c = 1/2$ at $T = 0$ by modelling the inter-pattern overlaps $\varphi_{1\mu} = \xi^1 \cdot \xi^\mu / N$ as Gaussian. A thermodynamic companion [2] recast this as a crossover between the retrieval free energy $f_{\text{ret}}(T)$ and a noise-floor energy $u_{\text{noise}}(\alpha)$, with the same $T \rightarrow 0$ value under the Gaussian. A second companion [3] replaced the Gaussian by the exact density of $\varphi_{1\mu}$ on the sphere; for the LSR kernel this raised the threshold from 0.25 to ≈ 0.347 .

This report applies the exact-density correction to LSE. The spurious partition sum $S_{\text{spur}} = \sum_{\mu \neq 1} \exp(\beta_{\text{net}} N \varphi_{1\mu})$ acquires an interior saddle at $\varphi^* = (\sqrt{5} - 1)/2$ with value $g_{\text{max}} = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774$. From $u_{\text{noise}}(\alpha) = 1 - (\alpha + g_{\text{max}})/\beta_{\text{net}}$ and the crossover condition $f_{\text{ret}}(T) = u_{\text{noise}}$, the boundary at $\beta_{\text{net}} = 1$ is $\alpha_c^{\text{sd}}(T) = 1 - g_{\text{max}} - f_{\text{ret}}(T)$, so $\alpha_c^{\text{sd}}(0) \approx 0.6226$, not 0.5. Transplanting the LSR support-edge derivation to LSE gives a different formula $\alpha_c^{\text{bd}}(T)$ that diverges at $T \rightarrow 0$; it is correct only for $\alpha < 0.241$ (where the saddle leaves the support).

The single-pattern spinodal $T_{\text{sp}}(\alpha)$, the temperature at which the retrieval basin loses local stability, lies above α_c^{sd} at $T > 0$ and meets it at the same endpoint $(1 - g_{\text{max}}, 0)$. In leading order $\alpha_c^{\text{sp}}(T) = \phi_{\text{eq}}(T) - g_{\text{max}}$, where $\phi_{\text{eq}}(T) = \frac{1}{2}(-T + \sqrt{T^2 + 4})$ is the basin minimum overlap.

Numerical status: a Monte Carlo heatmap at $N = 25$ over $\alpha \in [0.20, 0.70]$, $T \in [0.025, 0.500]$ from two MC variants (direct, semismart) and a cold-init spinodal probe at $N \in \{50, 100, 1000\}$ for $\alpha \in [0.20, 0.60]$. α_c^{sd} tracks the $N = 25$ residual at $\alpha \gtrsim 0.45$. The empirical spinodal anchors sit below α_c^{sp} across most of the range, except at $(\alpha, N) = (0.60, 1000)$ where the deviation is an NLO correction to the spinodal itself (verified by a Kramers budget calculation). A list of remaining tasks closes the report.

1 What we want

Model. A Dense Associative Memory [1] stores patterns $\xi^\mu \in \mathbb{R}^N$, $\mu = 1, \dots, M$, by encoding them as minima of an energy function $H : \mathbb{R}^N \rightarrow \mathbb{R}$. Retrieval of ξ^1 from a noisy cue is the Metropolis dynamics of $\mathbf{x}$ under H at external temperature T . Both patterns and chain lie on the N -sphere of radius $\sqrt{N}$: $|\xi^\mu|^2 = |\mathbf{x}|^2 = N$.

The log-sum-exp (LSE) kernel [4] defines

$$H_{\text{LSE}}(\mathbf{x}) = -\frac{1}{\beta_{\text{net}}} \ln \sum_{\mu=1}^M \exp(\beta_{\text{net}} \mathbf{x} \cdot \xi^\mu), \quad (1)$$

where β_{net} is the kernel sharpness (inverse variance) and has no relation to the external temperature. Throughout this report $\beta_{\text{net}} = 1$.

Overlaps and load. The chain–target and inter-pattern overlaps are

$$\varphi_1(\mathbf{x}) = \frac{\boldsymbol{\xi}^1 \cdot \mathbf{x}}{N}, \quad \varphi_{1\mu} = \frac{\boldsymbol{\xi}^1 \cdot \boldsymbol{\xi}^\mu}{N}, \quad (2)$$

both in $[-1, 1]$. Perfect retrieval is $\varphi_1 = 1$. The pattern count scales as

$$M = e^{\alpha N}, \quad \alpha = \frac{\ln M}{N}, \quad (3)$$

and α is the load. For uniformly distributed patterns on the sphere, $\varphi_{1\mu}$ has the exact density

$$f(\varphi) = \frac{(1 - \varphi^2)^{(N-3)/2}}{B((N-1)/2, 1/2)}, \quad \varphi \in [-1, 1], \quad (4)$$

with mean 0 and variance $1/N$.

Question. At $T = 0$, Ramsauer et al. proved that retrieval succeeds up to $\alpha_c = 1/2$, replacing $f(\varphi)$ by the Gaussian $\mathcal{N}(0, 1/N)$. The thermodynamic recast of the same argument [2] produces a smooth boundary $\alpha_c(T)$ at finite T , equal to $1/2$ at $T = 0$ under the same Gaussian. The companion LSR paper [3] showed that using (4) instead of the Gaussian shifts the LSR threshold from 0.25 to ≈ 0.347 . This report asks: what does the exact-density correction give for LSE?

2 Free energy framework

The boundary $\alpha_c(T)$ comes from a competition between two energies per spin: the retrieval free energy $f_{\text{ret}}(T)$ and the noise-floor energy $u_{\text{noise}}(\alpha)$. We derive both explicitly.

Retrieval free energy $f_{\text{ret}}(T)$. For a chain in the basin of $\boldsymbol{\xi}^1$ ($\varphi_1 \approx 1$, $\varphi_{1\mu} \approx 0$ for $\mu \neq 1$), the sum in (1) is dominated by $\mu = 1$:

$$\frac{H_{\text{LSE}}(\mathbf{x})}{N} \approx -\frac{1}{\beta_{\text{net}} N} \ln \exp(\beta_{\text{net}} N \varphi_1) = -\varphi_1. \quad (5)$$

We define the energy per spin relative to the retrieval baseline,

$$u(\varphi_1) = 1 + \frac{H_{\text{LSE}}}{N} = 1 - \varphi_1, \quad (6)$$

so $u(1) = 0$ at perfect retrieval and $u(0) = 1$ at zero overlap. The geometric entropy of the $(N-1)$ -sphere at fixed $\varphi_1 = \phi$ is, per spin,

$$s(\phi) = \frac{1}{2} \ln(1 - \phi^2), \quad (7)$$

negative on $(0, 1)$ [2]. The retrieval free energy at overlap ϕ is

$$f(\phi, T) = u(\phi) - T s(\phi) = (1 - \phi) - \frac{T}{2} \ln(1 - \phi^2). \quad (8)$$

Minimising in ϕ , $\partial f / \partial \phi = 0$ gives $T\phi = 1 - \phi^2$ and the basin overlap

$$\phi_{\text{eq}}(T) = \frac{1}{2}(-T + \sqrt{T^2 + 4}), \quad \phi_{\text{eq}}(0) = 1. \quad (9)$$

The retrieval free energy is the value at the minimum:

$$f_{\text{ret}}(T) = f(\phi_{\text{eq}}(T), T) = (1 - \phi_{\text{eq}}) - \frac{T}{2} \ln(1 - \phi_{\text{eq}}^2). \quad (10)$$

Noise-floor energy $u_{\text{noise}}(\alpha)$. A chain uncorrelated with ξ^1 has $\varphi_1 \approx 0$, so the $\mu = 1$ term in (1) is comparable to the others and the sum is dominated by the spurious patterns

$$S_{\text{spur}} = \sum_{\mu \neq 1} \exp(\beta_{\text{net}} N \varphi_{1\mu}). \quad (11)$$

For such a chain $H_{\text{LSE}}/N \approx -(\beta_{\text{net}} N)^{-1} \ln S_{\text{spur}}$, and by the convention (6),

$$u_{\text{noise}} = 1 - \frac{1}{\beta_{\text{net}} N} \ln S_{\text{spur}}. \quad (12)$$

Replacing the discrete sum by an integral over the density (4), with $M = e^{\alpha N}$,

$$S_{\text{spur}} \approx M \int_{-1}^1 f(\varphi) \exp(\beta_{\text{net}} N \varphi) d\varphi = e^{\alpha N} \int_{-1}^1 e^{N g(\varphi)} d\varphi, \quad (13)$$

where we have set the integrand exponent (per spin) to

$$g(\varphi) \equiv \frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}} \varphi, \quad (14)$$

using $f(\varphi) \propto \exp[(N/2) \ln(1 - \varphi^2)]$ at large N . Evaluating (13) by saddle in the large- N limit,

$$\frac{1}{N} \ln S_{\text{spur}} \approx \alpha + g_{\text{max}}, \quad g_{\text{max}} \equiv \max_{\varphi} g(\varphi). \quad (15)$$

Substituting into (12):

$$\boxed{u_{\text{noise}}(\alpha) = 1 - \frac{\alpha + g_{\text{max}}}{\beta_{\text{net}}}.} \quad (16)$$

Crossover. The two energies cross at $\alpha_c(T)$:

$$f_{\text{ret}}(T) = u_{\text{noise}}(\alpha_c) \implies \alpha_c(T) = \beta_{\text{net}} [1 - f_{\text{ret}}(T)] - g_{\text{max}}. \quad (17)$$

For $\beta_{\text{net}} = 1$:

$$\boxed{\alpha_c(T) = 1 - g_{\text{max}} - f_{\text{ret}}(T).} \quad (18)$$

The three predictions of Section 3 differ only in the value of g_{max} , determined by which $f(\varphi)$ and which integration range one substitutes into (13).

3 Three boundary formulas

Three choices of density and integration range, three values of g_{max} , three boundaries.

Gaussian density. Replace $f(\varphi)$ by the moment-matched Gaussian $\sqrt{N/2\pi} \exp(-N\varphi^2/2)$. The integrand exponent becomes $g_G(\varphi) = -\frac{1}{2}\varphi^2 + \beta_{\text{net}}\varphi$, monotone increasing on $[-1, 1]$ for $\beta_{\text{net}} = 1$. The integral is dominated by the upper cutoff, set by the typical maximum of M Gaussian samples: $M \cdot \Pr_G(\varphi_{1\mu} > \varphi_{\text{max}}^{(G)}) = 1$ gives $\varphi_{\text{max}}^{(G)}(\alpha) = \sqrt{2\alpha}$. At this edge, $g_G(\varphi_{\text{max}}^{(G)}) = \sqrt{2\alpha} - \alpha$, so by (15) and (16) $u_{\text{noise}}^{(G)} = 1 - \sqrt{2\alpha}$. The crossover (18) reads $1 - \sqrt{2\alpha_c^G} = f_{\text{ret}}(T)$, hence

$$\alpha_c^G(T) = \frac{1}{2} [1 - f_{\text{ret}}(T)]^2, \quad \alpha_c^G(0) = \frac{1}{2}. \quad (19)$$

Exact density, support edge. Keep the exact density (4) but restrict the integration to the support edge $\varphi_{\max}(\alpha)$, defined by $M \cdot \Pr_f(\varphi_{1\mu} > \varphi_{\max}) = 1$:

$$\varphi_{\max}(\alpha) = \sqrt{1 - e^{-2\alpha}}. \quad (20)$$

At this edge $g(\varphi_{\max}) = -\alpha + \varphi_{\max}$ (using $\frac{1}{2} \ln(1 - \varphi_{\max}^2) = -\alpha$), so $u_{\text{noise}}^{(\text{bd})} = 1 - \varphi_{\max}(\alpha)$. Solving (18):

$$\alpha_c^{\text{bd}}(T) = -\frac{1}{2} \ln[1 - (1 - f_{\text{ret}}(T))^2], \quad \alpha_c^{\text{bd}}(0) = +\infty. \quad (21)$$

This is the procedure that works for the LSR kernel, whose monotone $g(\varphi)$ has no interior saddle [3]. For LSE the procedure is correct only when $g(\varphi)$ is monotone on $[0, \varphi_{\max}(\alpha)]$; otherwise the interior saddle below dominates.

Exact density, interior saddle (this work). $g(\varphi)$ from (14) has an interior stationary point at $g'(\varphi) = -\varphi/(1 - \varphi^2) + \beta_{\text{net}} = 0$, equivalently $\beta_{\text{net}}(1 - \varphi^2) = \varphi$. For $\beta_{\text{net}} = 1$:

$$\varphi^{*2} + \varphi^* - 1 = 0 \implies \varphi^* = \frac{\sqrt{5}-1}{2} \approx 0.618 \text{ (golden ratio)}. \quad (22)$$

The identity $1 - \varphi^{*2} = \varphi^*$ gives the saddle value

$$g_{\max} = g(\varphi^*) = \frac{1}{2} \ln \varphi^* + \varphi^* \approx 0.3774. \quad (23)$$

Substituting into the master formula (18):

$$\boxed{\alpha_c^{\text{sd}}(T) = 1 - g_{\max} - f_{\text{ret}}(T) \approx 0.6226 - f_{\text{ret}}(T), \quad \alpha_c^{\text{sd}}(0) \approx 0.6226.} \quad (24)$$

Regime split. The interior saddle (22) dominates when it lies inside the integration range $[0, \varphi_{\max}(\alpha)]$, i.e. when $\varphi_{\max}(\alpha) \geq \varphi^*$. Solving $\sqrt{1 - e^{-2\alpha}} = \varphi^*$ gives the regime split at

$$\alpha_{\text{split}} = -\frac{1}{2} \ln(1 - \varphi^{*2}) = -\frac{1}{2} \ln \varphi^* \approx 0.241. \quad (25)$$

For $\alpha < \alpha_{\text{split}}$ the saddle lies outside support, $g(\varphi)$ is monotone on the available range, and α_c^{bd} applies. For $\alpha \geq \alpha_{\text{split}}$ the interior saddle dominates and α_c^{sd} applies. Our MC window $\alpha \in [0.20, 0.70]$ is almost entirely in the saddle regime.

Summary. Three boundaries from the master formula (18), differing only in $g_{\max}$. At $T = 0$, where $f_{\text{ret}}(0) = 0$: $\alpha_c^{\text{G}}(0) = 0.5$ (Gaussian, support-edge saddle), $\alpha_c^{\text{bd}}(0) = +\infty$ (exact density, support-edge saddle, divergent), $\alpha_c^{\text{sd}}(0) \approx 0.6226$ (exact density, interior golden-ratio saddle). Only α_c^{sd} is correct for LSE at $\alpha \geq \alpha_{\text{split}}$.

4 Spinodal of the single-pattern phase

The boundary $\alpha_c^{\text{sd}}(T)$ marks the global free-energy crossover between the single-pattern phase and the saddle-dominated phase. Above the crossover, the saddle phase wins thermodynamically but the single-pattern minimum can persist as a metastable local minimum until its own marginal stability fails. We call the temperature at which the local minimum disappears $T_{\text{sp}}(\alpha)$. At fixed α , $T_{\text{sp}}(\alpha) \geq T_c^{\text{sd}}(\alpha)$.

A chain initialised at ξ^1 (cold init) sits inside the single-pattern basin. If the barrier between the basin and the saddle phase is large, the chain cannot escape within MC time and traces $T_{\text{sp}}(\alpha)$ rather than $T_c^{\text{sd}}(\alpha)$.

Leading-order formula. At fixed $\varphi_1 = \xi^1 \cdot \mathbf{x}/N$ the spurious sum evaluates by saddle to $e^{N(\alpha+g_{\max})}$, independent of $\mathbf{x}$ to leading order. The energy per spin is then $h(\varphi_1) = \max(\varphi_1, \alpha + g_{\max})$. With the $(N-1)$ -sphere entropy $s(\varphi_1) = \frac{1}{2} \ln(1 - \varphi_1^2)$,

$$\frac{F(\varphi_1)}{N} = -\max(\varphi_1, \alpha + g_{\max}) - \frac{T}{2} \ln(1 - \varphi_1^2). \quad (26)$$

On the single-pattern branch $\varphi_1 > \alpha + g_{\max}$, the condition $dF/d\varphi_1 = 0$ recovers $\varphi_1 = \phi_{\text{eq}}(T)$. The local minimum exists only while $\phi_{\text{eq}}(T) \geq \alpha + g_{\max}$. Saturating the inequality:

$$\boxed{\alpha_c^{\text{sp}}(T) = \phi_{\text{eq}}(T) - g_{\max}, \quad T_{\text{sp}}(\alpha) = \frac{1 - (\alpha + g_{\max})^2}{\alpha + g_{\max}}.} \quad (27)$$

At $T = 0$ both α_c^{sd} and α_c^{sp} pass through $(\alpha, T) = (1 - g_{\max}, 0) \approx (0.6226, 0)$.

What is robust, what is not. The endpoint $(1 - g_{\max}, 0)$ is robust against subleading $1/N$ corrections and against soft-cusp rounding of $h(\varphi_1)$ near $\varphi_1 = \alpha + g_{\max}$: both shift the location of the local minimum at fixed T by $O(\log N/N)$ but do not move the spinodal *curve* to that order. The *slope* of $\alpha_c^{\text{sp}}(T)$ at $T > 0$ depends on those refinements; (27) is a guide, not a verified prediction.

Exact LO barrier. The LO free energy (26) is piecewise analytic, so the barrier between the retrieval minimum and the cusp is the exact difference of F/N at the two points. Writing $\xi = \phi_{\text{eq}}(T) - (\alpha + g_{\max})$,

$$\frac{\Delta F(\alpha, T)}{N} = \xi + \frac{T}{2} \ln \frac{1 - \phi_{\text{eq}}(T)^2}{1 - (\alpha + g_{\max})^2}. \quad (28)$$

ΔF vanishes at the spinodal ($\xi = 0$) and is positive below it. No Taylor expansion of F is needed; the LO free energy is itself the leading-order object, exact within that order.

Where the LO free energy is wrong. Two effects are missing from (26), both visible at finite N .

Cusp rounding. The kink in $h(\varphi_1)$ at $\varphi_1 = \alpha + g_{\max}$ is the $N \rightarrow \infty$ limit of a smooth crossover. The total partition sum inside the LSE logarithm is $e^{\beta_{\text{net}} N \varphi_1} + S_{\text{spur}}$, the retrieval term plus the spurious sum. At leading order the spurious sum is $S_{\text{spur}} \approx e^{N(\alpha+g_{\max})}$ (15), and the log of a sum of two exponentials collapses to the maximum of their arguments:

$$\frac{1}{\beta_{\text{net}} N} \ln[e^{\beta_{\text{net}} N \varphi_1} + e^{N(\alpha+g_{\max})}] \xrightarrow{N \rightarrow \infty} \max(\varphi_1, \alpha + g_{\max}). \quad (29)$$

This collapse, used to obtain (26), is the source of the kink. At finite N the log-sum is smooth: the two terms are comparable when $|\varphi_1 - (\alpha + g_{\max})|$ is of order $1/(\beta_{\text{net}} N)$, and inside that window the log-sum sits above the maximum by an amount of order $1/(\beta_{\text{net}} N)$. The kink is therefore rounded over a window of width

$$\Delta\varphi_1 \sim 1/(\beta_{\text{net}} N). \quad (30)$$

The LO barrier (28) computes the height of the unrounded kink and is unreliable when the retrieval basin sits inside or near the rounding window, i.e. when $\xi = \phi_{\text{eq}}(T) - (\alpha + g_{\max})$ is comparable to $1/(\beta_{\text{net}} N)$ or smaller.

The basin offset at fixed relative temperature satisfies $\xi \approx \frac{1}{2} r T_{\text{sp}}^{\text{LO}}(\alpha)$ for $T = (1 - r) T_{\text{sp}}^{\text{LO}}(\alpha)$. Near the cusp $1 - g_{\max} \approx 0.6226$ the LO spinodal $T_{\text{sp}}^{\text{LO}}(\alpha)$ is small, so even at modest r the basin offset ξ falls inside the rounding window. Concretely, at $\alpha = 0.60$, $N = 1000$ the window is $1/N = 0.001$; at $T = 0.95 T_{\text{sp}}^{\text{LO}}$ the LO offset is $\xi \approx 0.0012$, comparable to the window. The LO

formula therefore misrepresents the basin shape there, and the true spinodal shifts above $T_{\text{sp}}^{\text{LO}}$ by an NLO correction. At $\alpha = 0.55$, $N = 1000$ the same calculation gives $\xi \approx 0.0038$, four times the rounding window, and the LO formula is accurate (consistent with the empirical $T_{\text{sp}}^{\text{MC}} = 0.150$ versus $T_{\text{sp}}^{\text{LO}} = 0.151$).

Extreme-value regime. The replacement of S_{spur} by its mean (15) is reliable only when $\text{Var}(S_{\text{spur}}) \ll \langle S_{\text{spur}} \rangle^2$. The mean is $\langle S_{\text{spur}} \rangle \approx \exp[N(\alpha + g_{\text{max}}(\beta_{\text{net}}))]$. The second moment splits into a diagonal part (terms with $\mu = \nu$) and an off-diagonal part (terms with $\mu \neq \nu$):

$$\langle S_{\text{spur}}^2 \rangle = M \langle e^{2\beta_{\text{net}}N\varphi} \rangle + M(M-1) \langle e^{\beta_{\text{net}}N\varphi} \rangle^2. \quad (31)$$

The diagonal part is the same saddle calculation as (15) with the kernel inverse variance β_{net} replaced by $2\beta_{\text{net}}$: $M \langle e^{2\beta_{\text{net}}N\varphi} \rangle \approx \exp[N(\alpha + g_{\text{max}}(2\beta_{\text{net}}))]$. The off-diagonal part is $\langle S_{\text{spur}} \rangle^2$. Subtracting:

$$\frac{\text{Var}(S_{\text{spur}})}{\langle S_{\text{spur}} \rangle^2} \approx \exp[N(g_{\text{max}}(2\beta_{\text{net}}) - 2g_{\text{max}}(\beta_{\text{net}}) - \alpha)]. \quad (32)$$

The mean replacement fails when the exponent is positive, i.e. for $\alpha < \alpha_{\text{var}}$ with

$$\alpha_{\text{var}} \equiv g_{\text{max}}(2\beta_{\text{net}}) - 2g_{\text{max}}(\beta_{\text{net}}). \quad (33)$$

For $\beta_{\text{net}} = 1$ the saddle at doubled sharpness solves $2(1 - \varphi^2) = \varphi$, giving $\varphi^{**} = (-1 + \sqrt{17})/4 \approx 0.781$ and $g_{\text{max}}(2) = \frac{1}{2} \ln(1 - \varphi^{**2}) + 2\varphi^{**} \approx 1.091$. With $g_{\text{max}}(1) \approx 0.3774$,

$$\alpha_{\text{var}} \approx 1.091 - 2(0.3774) \approx 0.336. \quad (34)$$

Below α_{var} the typical S_{spur} is dominated by the single largest overlap rather than the saddle value. That maximum overlap is $\varphi_{\text{max}}(\alpha) = \sqrt{1 - e^{-2\alpha}}$ (20), so the effective cusp position moves from $\alpha + g_{\text{max}}$ towards $\max(\alpha + g_{\text{max}}, \varphi_{\text{max}}(\alpha))$. This is the same crossover as the regime split (25) at $\alpha_{\text{split}} \approx 0.241$. For $\alpha \in [0.20, 0.70]$ the saddle value dominates the cusp throughout, and this correction is small in our window.

Kramers budget under LO. The Arrhenius escape time is $\tau_{\text{esc}} \approx \exp(N \Delta F/T)$. With detection target $T_{\text{dep}} = (1 - r) T_{\text{sp}}^{\text{LO}}(\alpha)$ the required sample budget is

$$N_{\text{SAMP}}^{\text{req}}(\alpha, N, r) = \exp\left(\frac{N \Delta F(\alpha, T_{\text{dep}})}{T_{\text{dep}}}\right), \quad (35)$$

with ΔF from (28) evaluated at $T = T_{\text{dep}}$. Figure 1 plots (35) on a grid; the script `plot_budget_vs_alpha.jl` reproduces the curves.

Two conclusions. (i) At $r = 5\%$, $N_{\text{SAMP}}^{\text{req}} \leq 2$ across the whole (α, N) range, three orders of magnitude below our budget of 2,000. (ii) Inverting (35) for r at fixed $N_{\text{SAMP}} = 2,000$ gives a resolution of $r \approx 17\%$ at $\alpha = 0.60$, $N = 1000$ (≈ 0.008 in T). The empirical gap at that cell is 0.035, four times the budget resolution, so it cannot be a finite-MC-time effect.

Summary: which formula in which α interval. The thermodynamic boundary $\alpha_c(T)$ and the spinodal $T_{\text{sp}}(\alpha)$ are piecewise across the α axis. Two thresholds split the axis: $\alpha_{\text{split}} \approx 0.241$ (25), where the interior saddle φ^* enters the support, and $\alpha_{\text{var}} \approx 0.336$ (34), where S_{spur} begins to self-average. In the master formula $\alpha_c(T) = 1 - g_{\text{max}} - f_{\text{ret}}(T)$, the operative value of g_{max} depends on the α range:

- $\alpha < \alpha_{\text{split}} \approx 0.241$ (boundary regime). The interior saddle of $g(\varphi) = \frac{1}{2} \ln(1 - \varphi^2) + \beta_{\text{net}}\varphi$ lies outside the natural integration range $[0, \varphi_{\text{max}}(\alpha)]$; the integrand is monotone and dominated by the support edge $\varphi_{\text{max}}(\alpha) = \sqrt{1 - e^{-2\alpha}}$. The correct boundary is $\alpha_c^{\text{bd}}(T)$ (21). Self-averaging fails here ($\alpha < \alpha_{\text{var}}$), so even $\alpha_c^{\text{bd}}(T)$ is only a typical-value statement, not a sharp transition. The Gaussian formula $\alpha_c^{\text{G}}(T)$ (19) agrees with α_c^{bd} to leading order in α but uses an inflated support edge $\sqrt{2\alpha}$ instead of $\varphi_{\text{max}}(\alpha)$.

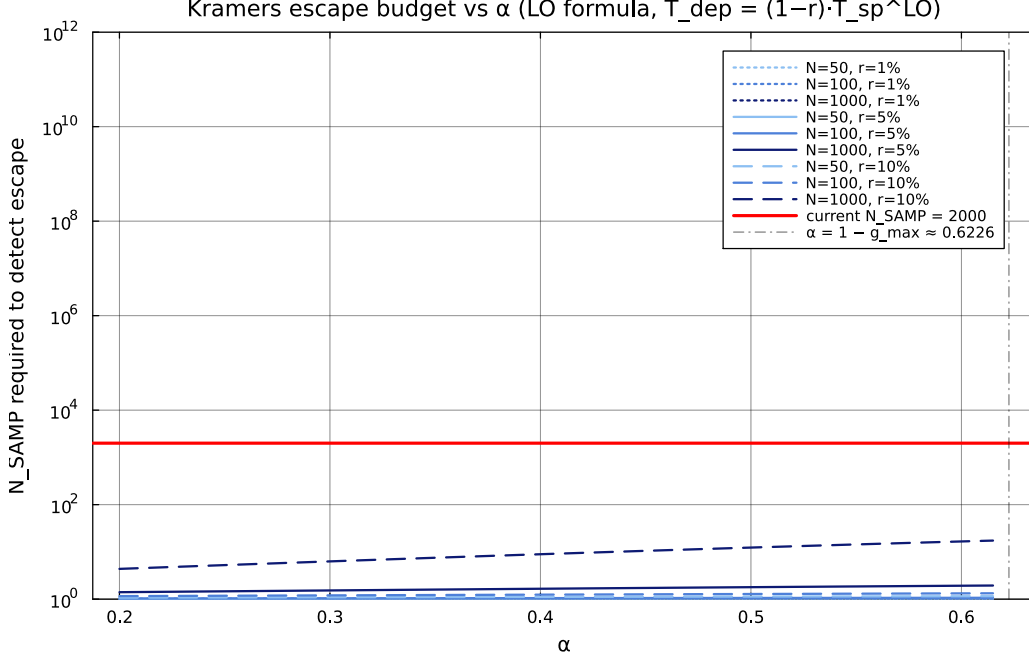


Figure 1: Kramers escape budget (35) versus α for $N \in \{50, 100, 1000\}$ at relative offsets $r \in \{1, 5, 10\}\%$ from $T_{\text{sp}}^{\text{LO}}(\alpha)$. Red line: current MC budget $N_{\text{SAMP}} = 2,000$. All curves sit far below the red line, so the budget is sufficient at every (α, N) in the window. Dash-dot: cusp $\alpha = 1 - g_{\text{max}} \approx 0.6226$.

- $\alpha_{\text{split}} \leq \alpha < \alpha_{\text{var}} \approx 0.336$ (saddle-dominated mean, non-self-averaging). The interior saddle at $\varphi^* = (\sqrt{5} - 1)/2$ lies inside the support and the saddle formula $\alpha_c^{\text{sd}}(T)$ (24) gives the correct *mean* boundary. Sample-to-sample fluctuations in S_{spur} are still exponentially large, so the boundary position fluctuates from one disorder realization to the next; only its mean is set by α_c^{sd} .
- $\alpha \geq \alpha_{\text{var}} \approx 0.336$ (saddle-dominated, self-averaging). $\alpha_c^{\text{sd}}(T)$ is the correct boundary and the realization-to-realization spread vanishes exponentially with N . This is the regime where (24) should match Monte Carlo without finite- N scatter in α .

Spinodal piecewise structure. The single-pattern spinodal $\alpha_c^{\text{sp}}(T) = \phi_{\text{eq}}(T) - g_{\text{max}}$ (27) uses the same g_{max} as α_c^{sd} , so it carries the same regime structure. In the saddle regime ($\alpha > \alpha_{\text{split}}$) it sits above α_c^{sd} at $T > 0$ and they meet at the endpoint $(1 - g_{\text{max}}, 0) \approx (0.6226, 0)$. Below α_{split} the LO derivation of α_c^{sp} uses an incorrect cusp position and (27) does not apply; whether a single-pattern spinodal exists in the boundary regime is a separate question we do not address. Within the saddle regime, the LO formula (27) is accurate at fixed α as long as the basin offset ξ exceeds the cusp-rounding window (30), i.e. for T not too close to $T_{\text{sp}}^{\text{LO}}$ and for α not too close to the cusp $1 - g_{\text{max}}$.

5 Monte Carlo methods

All chains run on the $\sqrt{N}$ -sphere with $\beta_{\text{net}} = 1$, Metropolis updates, step size $\sigma(T) = 2.4T/\sqrt{N}$. Three variants differ in how they handle the $M = e^{\alpha N}$ patterns.

Direct (all patterns). All M patterns are generated and stored explicitly. The chain evaluates (1) over all of them at every step. Memory cost is $O(NM)$; at $\alpha = 0.7$ and $N = 25$ this is $M \approx 4 \times 10^7$ patterns, near the limit of a single GPU with 40 GB. We use direct MC

for the heatmap reference at $N = 25$ across $\alpha \in [0.20, 0.70]$. Direct MC is impossible at the $N \in \{50, 100, 1000\}$ used for the spinodal probe: $e^{0.6 \cdot 1000} \approx 10^{260}$.

Semismart (saddle-band, anchored at ξ^1). Only patterns with overlap $\varphi_{1\mu} \geq \varphi_{\text{keep}}$ to the target ξ^1 are stored explicitly. φ_{keep} is chosen so that the expected number above the cutoff equals a budget K_{target} (we use $K_{\text{target}} = 10^4$); the actual count K_{kept} is drawn from $\text{Poisson}(M \cdot p)$ with $p = \Pr(\varphi > \varphi_{\text{keep}})$, and the K_{kept} overlap values are drawn from a truncated Beta distribution. The contribution of the discarded tail is the analytic constant

$$\log C_{\text{bulk}} = \alpha N + \ln \int_{-1}^{\varphi_{\text{keep}}} f(\varphi) e^{\beta_{\text{net}} N \varphi} d\varphi, \quad (36)$$

added inside the LSE log-sum at every step.

The retained set is quenched: each disorder sample fixes the retained patterns once and reuses them for the chain. At cold init the chain stays near ξ^1 , so the kept patterns dominate the local energetics of escape; the bulk acts as a smooth background.

Smart (top-K, dynamic anchor). Same as semismart, but the retained set is refreshed every N_{REFRESH} steps based on the chain's current position. Used in the LSE Nramp scripts where the chain wanders far from ξ^1 . The spinodal probe is a cold-init experiment with negligible chain wandering, so we use semismart, not smart.

What each variant measures. Direct MC measures the true thermodynamics at fixed N . Semismart MC faithfully measures the local stability of the retrieval well around ξ^1 (the spinodal) and reproduces the saddle thermodynamics through C_{bulk} . Smart MC tracks the chain across the energy landscape and is needed when the chain leaves the single-pattern basin and explores the saddle phase.

Top-K vs. saddle patterns. Two pattern populations matter in LSE thermodynamics. Patterns with high $\varphi_{1\mu}$ (top-K) are the strongest competitors at the retrieval well: they set the local curvature of H at $\mathbf{x} = \xi^1$ and therefore the spinodal. The saddle constellation patterns at $\varphi \approx 0.618$ have random (typically near-zero) overlap with ξ^1 . They do not enter the local stability of the retrieval well; they enter α_c^{sd} through the partition-function competition with $f_{\text{ret}}(T)$. Semismart MC samples top-K and is the right tool for the spinodal. The saddle thermodynamics is added analytically through C_{bulk} , not measured by the chain.

6 What we have

Two datasets so far:

1. Heatmap at $N = 25$, $\alpha \in [0.20, 0.70]$ step 0.01, $T \in [0.025, 0.500]$ on 40 points, using direct MC and semismart MC on the same grid. Cell value: $\langle \phi \rangle - \phi_{\text{eq}}(T)$ averaged over disorder. See Figure 2.
2. Spinodal probe: semismart MC at $\alpha \in \{0.20, 0.25, \dots, 0.60\}$, $N \in \{50, 100, 1000\}$, cold init only, per- α fine T grids, 8 disorders, ceiling at $T = 0.5$ (cells with $\langle \phi \rangle / \phi_{\text{eq}} > 0.9$ at $T = 0.5$ are reported as NaN). See Table 1.

Heatmap. Figure 2 shows three things. First, the Gaussian curve α_c^{G} sits deep inside the spin-glass region for the whole range. Second, the boundary transplant α_c^{bd} sits far outside the empirical transition. Third, the saddle curve α_c^{sd} tracks the empirical transition for $\alpha \gtrsim 0.45$ at $N = 25$. At $\alpha < 0.45$ the empirical transition lies at lower T than α_c^{sd} predicts.

E basin stability + 4 analytical $\alpha_c(T)$ (Nramp $\oplus$ N=25, multi-source)

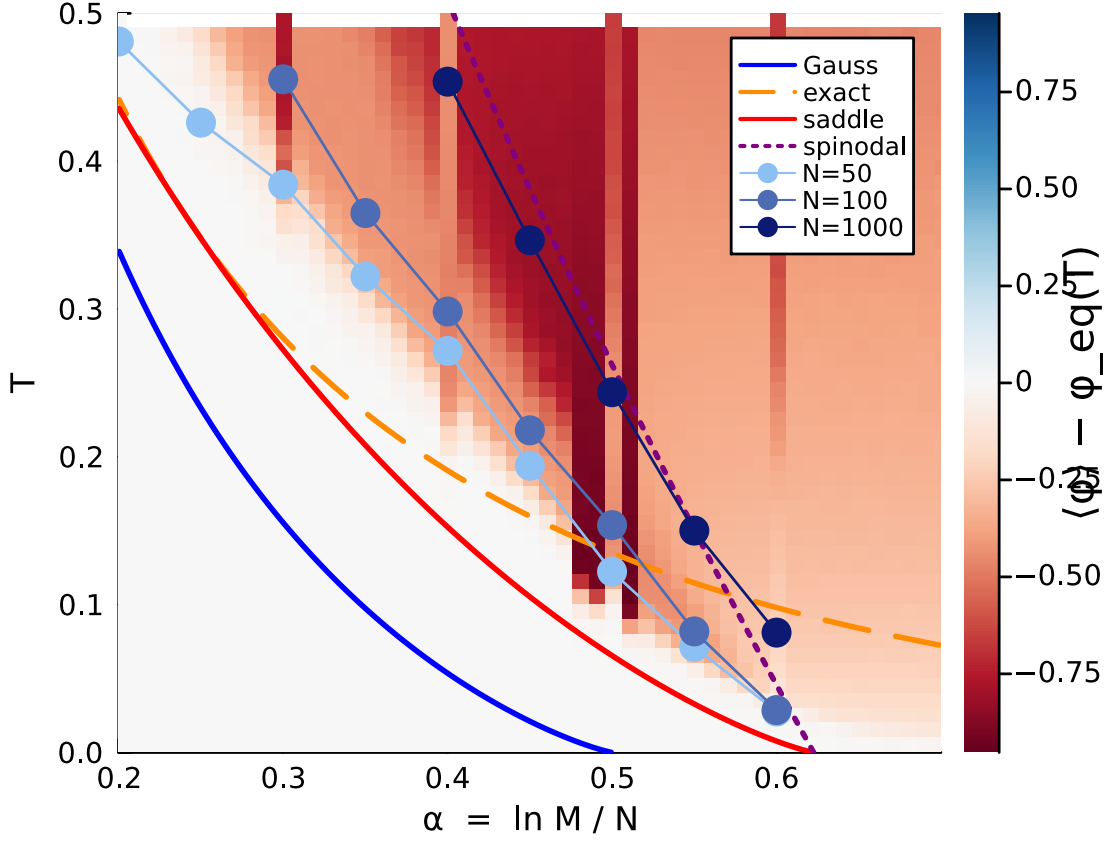


Figure 2: Residual $\langle \phi \rangle - \phi_{eq}(T)$ from direct and semismart MC at $N = 25$, with the three boundary curves (19), (21), (24) overlaid. Filled circles: empirical cold-init spinodal anchors $T_{sp}^{MC}(\alpha, N)$ at $N \in \{50, 100, 1000\}$ (light to dark blue). Violet dashed line: leading-order spinodal $\alpha_c^{sp}(T)$ from (27).

Spinodal anchors. Table 1 reports the cold-init boundary T_{sp}^{MC} detected as the temperature at which $\langle \phi \rangle$ first drops below $0.9 \phi_{eq}(T)$. The leading-order prediction (27) is shown for comparison.

Three observations.

(1) For fixed α , $T_{sp}^{MC}(\alpha, N)$ increases monotonically with N toward $T_{sp}^{LO}(\alpha)$. At $\alpha = 0.55$, $N = 1000$ the empirical and LO values agree.

(2) For $\alpha \in [0.30, 0.50]$, $T_{sp}^{MC}(\alpha, 1000)$ sits below $T_{sp}^{LO}(\alpha)$. This is the expected sign of sub-leading $1/N$ corrections to the spinodal.

(3) At $\alpha = 0.60$, $N = 1000$, $T_{sp}^{MC} = 0.081$ sits *above* $T_{sp}^{LO} = 0.046$. The budget analysis (Figure 1, Section 4) rules out a finite-MC-time floor: the LO Kramers shift at this cell is at most 0.008, four times smaller than the observed gap. The deviation is an NLO correction to the spinodal location itself. In this regime $\xi \lesssim \log N/N$ at $T = T_{sp}^{LO}$, so the parabolic LO barrier vanishes and the cusp-rounded NLO physics dominates.

Hot init not yet run. The hot-init column T_{sp}^{hot} in `spinodal_summary.csv` is uniformly NaN: the run used cold init only. The hot-init boundary identifies the thermodynamic $T_c^{sd,MC}(\alpha)$ and, paired with cold init, delimits the metastability band. This is the most important missing piece.

Table 1: Empirical spinodal $T_{\text{sp}}^{\text{MC}}$ from cold-init semismart MC, compared to leading-order $T_{\text{sp}}^{\text{LO}}$ from (27). NaN means no departure detected by the ceiling $T = 0.5$.

α	$N = 50$	$N = 100$	$N = 1000$	$T_{\text{sp}}^{\text{LO}}$
0.20	0.481	NaN	NaN	1.154
0.25	0.426	NaN	NaN	0.966
0.30	0.384	0.455	NaN	0.799
0.35	0.322	0.365	NaN	0.647
0.40	0.272	0.298	0.454	0.509
0.45	0.194	0.218	0.347	0.381
0.50	0.122	0.154	0.244	0.262
0.55	0.071	0.082	0.150	0.151
0.60	0.028	0.029	0.081	0.046

7 What is left

The status of each task: *done, in progress, open*.

Numerical tasks.

- Hot-init spinodal probe** (*open*). Add hot init (uniform sphere start) at the same (α, N, T) grid as the cold-init run. Cold and hot boundaries together delimit the metastability band; cold $\rightarrow \alpha_c^{\text{sp}}$, hot $\rightarrow \alpha_c^{\text{sd}}$.
- Finite-MC-time test at $\alpha = 0.60$, $N = 1000$** (*closed by analysis*). Settled by Figure 1: LO Kramers shift at this cell is ≤ 0.008 , far below the observed 0.035 gap. The deviation is NLO, not finite-MC-time. An MC re-run with larger N_{SAMP} would not move the empirical $T_{\text{sp}}^{\text{MC}}$.
- Extension of spinodal anchors toward the cusp** (*open*). Add $\alpha \in \{0.605, 0.610, 0.615, 0.620, 0.625\}$ with extended N_{SAMP} . Goal: confirm that the empirical curve bends to $T = 0$ at $\alpha = 1 - g_{\text{max}} \approx 0.6226$.
- Smart-MC heatmap at larger N** (*open*). The heatmap is at $N = 25$. The saddle-band-keeping semismart code runs at $N \in \{50, 100, 1000\}$. An $N = 50$ heatmap over $\alpha \in [0.20, 0.70]$ on the same T grid would show whether the discrepancy at $\alpha < 0.45$ at $N = 25$ shrinks with N (thermodynamic correction) or persists (kinetic effect).
- Chain-length scan** (*open*). Fix (α, T) inside the small- α discrepancy region. Sweep N_{chain} over factor-of-four steps from 2^{14} to 2^{22} . Drift of the residual toward α_c^{sd} as $\log N_{\text{chain}}$ grows = kinetic escape. Flat residual = thermodynamic finite- N correction.
- Direct measurement of the Kramers barrier** (*open*). Long chain inside the basin gives the histogram $P(\varphi_1)$ and hence $V_{\text{eff}}(\varphi_1) = -T \ln P(\varphi_1)$. Read ΔF from the histogram and predict the escape time by Arrhenius.

Analytical tasks.

- Subleading $1/N$ corrections to α_c^{sp}** (*open*). Expand (26) beyond leading order. Two sources, both discussed in Section 4: cusp rounding at scale $1/(\beta_{\text{net}} N)$ in φ_1 , and the extreme-value correction below the self-averaging threshold $\alpha_{\text{var}} \approx 0.336$. Goal: quantitative prediction of $T_{\text{sp}}(\alpha, N)$ to compare against the empirical anchors at $N \in \{50, 100, 1000\}$.

8. **Spinodal formula from the exact-density free energy** (*open*). Re-derive (27) from the same exact-density partition function that yields α_c^{sd} , without the leading-order saddle approximation. Goal: close the gap between the empirical $T_{\text{sp}}^{\text{MC}}$ at $\alpha = 0.30$, $N = 1000$ (NaN — no departure detected) and the LO value $T_{\text{sp}}^{\text{LO}} = 0.80$.
9. **LSE Kramers barrier** (*open*). Derive $\Delta F(\alpha, T)$ for escape from the LSE retrieval well in closed form. The LSE landscape is smooth (no LSR support cutoff), so the escape mechanism differs from the LSR centroid hopping of [3]. Goal: analytic Arrhenius form to compare with task 6.

Writing tasks.

10. **Figure of the empirical anchors vs. LO** (*done*: included in Figure 2).
11. **Caption pass** (*open*). Figures and tables to be rewritten to match the report style.
12. **References** (*open*). `refs.bib` contains the four citations used here; check that all keys resolve in the final PDF.

Decision points for the user. Two choices shape the next sprint. First: do we treat the small- α discrepancy as kinetic or thermodynamic? Tasks 4–6 discriminate. Second: does the paper claim $\alpha_c^{\text{sd}}(0) = 1 - g_{\text{max}} \approx 0.6226$ as the corrected Ramsauer threshold, or does it remain in the present descriptive form (“saddle dominates; transplant is wrong; here is the curve”)? The first claim is stronger and would benefit from task 4 first.

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